

Case study 3- Software Architecture of a System Designed for Churn Analysis and Customer Retention in Banking Industry (AI Generated Architecture by Microsoft Copilot)

1. System summary

The system is a generic software solution for churn analysis and customer retention in the banking industry. Its primary objectives are to:

- **Support customer recognition and profiling** based on transaction history, spending habits, and product usage.
- **Provide product recommendations and next best actions** optimized for both company revenue and customer satisfaction.
- **Enable churn analysis and prediction** using supervised learning models over historical customer data.
- **Manage machine learning models and monitoring** including training, deployment, performance measurement, and reporting.

The overall scope includes data preprocessing, model management, prediction and optimization services, recommendation and action services, and monitoring/reporting for banking organizations.

2. Assumptions

A-001–A-003 are given in the SRS and treated as requirements context.

Additional explicit architectural assumptions (beyond SRS assumptions):

1. **Data access assumption:** Banking transaction datasets are accessible via a stable data source interface (but no specific technology is assumed).
2. **Client interaction assumption:** Client applications and action services interact with the system via well-defined programming interfaces (APIs) as implied by FR-022.
3. **User interface assumption:** A single logical UI layer serves all human users (model creators/loaders, analysts, business users) even if physically realized as multiple screens.
4. **Deployment assumption:** The system is deployed in a multi-node environment (application nodes and data nodes) but no specific infrastructure technology is assumed.

3. Selected architecture style

Selected style: Layered, modular service-oriented architecture (logical layers with domain-focused components).

Justification

- **Clear separation of concerns:**
 - Data preprocessing, model management, prediction/recommendation, and monitoring/reporting are distinct functional areas (FR-030–FR-036, FR-007–FR-012, FR-022–FR-027).
- **Maintainability and modifiability:**
 - NFR-004–NFR-007, NFR-008–NFR-010 require measurable performance, easy comparison of predictions vs actuals, and continued functionality despite lower-level technology changes—best supported by layered, modular components with clear interfaces.
- **Flexibility with libraries and technologies:**
 - NFR-009–NFR-010 emphasize flexibility with open source technologies and new languages; a service-oriented, interface-driven design allows swapping implementations without changing higher layers.
- **Single system for training, prediction, and monitoring:**
 - NFR-005 requires the same system to handle training, prediction, and monitoring, which fits a layered architecture with shared domain services and a unified model management layer.

Other styles (e.g., microservices, event-driven) could be appropriate, but the SRS does not explicitly demand fine-grained distribution or specific event semantics. A logical layered architecture remains faithful to the SRS while leaving room for later physical decomposition.

4. Architectural drivers

4.1 Functional drivers

Key functional requirements shaping the architecture:

- **Customer profiling and recognition:** FR-001, FR-002 (profiling based on product use, spending habits).
- **Data provisioning for models:** FR-003, FR-004 (daily and streaming data for training, scoring, optimization).
- **Prediction and optimization services:** FR-005, FR-006, FR-017, FR-018, FR-019, FR-020, FR-026–FR-028 (churn prediction, customer tendency models, automatic prediction models, multi-model churn prediction).
- **Recommendation and next best actions:** FR-021–FR-025, FR-022–FR-023 (ranked recommendations, sequenced actions, next best action).
- **Data preprocessing:** FR-030–FR-031 (normalization, outlier handling, missing data imputation, feature selection, dimension reduction).
- **Model management:** FR-007–FR-009, FR-032–FR-036 (creation, implementation, training, activation/deactivation, metadata, analytics, reporting).
- **Monitoring and reporting:** FR-010–FR-014, FR-029, FR-034–FR-036 (performance measurement, comparison with actuals, optimization monitoring, model lists and usage analytics).

4.2 Quality attribute drivers

Architecturally significant non-functional requirements:

- **Performance (NFR-001):**
 - “Prediction and optimization results whenever needed” drives the need for efficient prediction and optimization services, separation of online scoring from heavy training, and caching/efficient data access at the logical level.
- **Usability (NFR-002–NFR-003):**
 - Easy model creation/implementation and a UI for loading models require a dedicated user interface layer and model management workflows.
- **Maintainability (NFR-004–NFR-007, NFR-010):**
 - Measurable performance, easy comparison of predictions vs actuals, monitoring optimization, and flexibility with updated libraries drive explicit monitoring, logging, and metadata components.
- **Modifiability and flexibility (NFR-008–NFR-009):**
 - Continued functionality despite lower-level technology changes and flexibility with open source technologies motivate strong abstraction boundaries (interfaces) between layers and components.

These drivers lead to a modular architecture with clear interfaces for data access, model management, prediction/recommendation services, and monitoring/reporting.

5. Requirements-to-architecture traceability

Requirement ID	Requirement	Architectural Component(s)	Justification
FR-001	Profile customers with similar/different product use and history.	Customer Profiling Service, Data Access Layer	Profiling logic uses transaction data and product usage to build profiles.
FR-002	Create customer profiles based on spending habits and product use.	Customer Profiling Service	Component responsible for deriving profiles from spending and usage patterns.
FR-003	Provide necessary daily data for model training, scoring, optimization.	Data Provisioning Service, Data Access Layer	Daily batch data extraction and preparation for models.
FR-004	Provide necessary streaming data for model training, scoring, optimization.	Streaming Data Provisioning Service	Handles continuous data feeds for online training/scoring.
FR-005	Provide prediction results whenever needed.	Prediction Service, API Gateway	Prediction service exposes interfaces for client applications.
FR-006	Provide optimization results whenever needed.	Optimization Service	Computes optimized recommendations/actions.
FR-007	Support creation of machine learning models.	Model Management Service, UI Layer	UI and backend workflows for model creation.

FR-008	Support implementation of machine learning models.	Model Deployment Service	Handles model implementation/activation in production.
FR-009	Provide UI for users who load models.	Model Management UI	Dedicated interface for model loaders.
FR-010	Enable measurement of performance of implemented models.	Monitoring & Analytics Service	Collects metrics and performance data.
FR-011	Support both model training and prediction.	Training Service, Prediction Service	Separate components for training and prediction under same system.
FR-012	Monitor model performance.	Monitoring & Analytics Service	Continuous performance monitoring.
FR-013	Enable comparison of predictions with actual results.	Monitoring & Analytics Service, Data Access Layer	Stores predictions and actuals for comparison.
FR-014	Enable users to monitor optimization performance.	Optimization Monitoring Module	Tracks optimization outcomes and exposes reports.
FR-015	Include newly developed models.	Model Registry	Stores and manages newly added models.
FR-016	Manage newly added modules.	Module Management Component	Handles registration and lifecycle of modules.
FR-017	Provide customer tendency models including probability of purchasing/using products.	Tendency Modeling Service	Builds and serves tendency models.
FR-018	Create automatic models for predictions.	Auto-Model Generation Service	Automates model creation based on data.
FR-019	Manage prediction models.	Model Registry, Model Management Service	Lifecycle management of prediction models.
FR-020	Monitor model performances.	Monitoring & Analytics Service	Performance dashboards and metrics.
FR-021	Determine best possible actions in proper order to prevent attrition.	Action Recommendation Engine	Computes ordered action sequences.
FR-022	Provide programming interface to recommend products/services to client apps.	API Gateway, Recommendation Service	Exposes recommendation APIs.

FR-023	Provide ranked recommendations for a given customer ID.	Recommendation Service	Returns ranked list of products/services.
FR-024	Recommend next best action for a given customer.	Action Recommendation Engine	Computes single next best action.
FR-025	Provide sequenced actions for a given customer ID.	Action Recommendation Engine	Returns ordered action list.
FR-026	Provide functions to conduct churn analysis.	Churn Analysis Service	Performs analysis over historical data.
FR-027	Provide churn prediction for given customer ID(s) and selected model.	Churn Prediction Service	Uses selected model to score customers.
FR-028	Utilize more than one model for churn prediction.	Churn Ensemble Service	Combines multiple models' outputs.
FR-029	Reporting functions listing existing supervised learning models in use.	Model Reporting Module	Generates model usage reports.
FR-030	Data preprocessing steps to enable ML-ready dataset.	Data Preprocessing Service	Prepares datasets for ML.
FR-031	Functions for normalization, outlier detection, missing data imputation, feature selection, dimension reduction.	Data Preprocessing Service	Implements preprocessing operations.
FR-032	Enable management of ML models.	Model Management Service	Central model lifecycle management.
FR-033	Functions for training, serializing, deserializing, activating, deactivating models.	Training Service, Model Deployment Service, Model Registry	Handles training and lifecycle operations.
FR-034	Report list and status of existing models.	Model Reporting Module	Exposes model list and status.
FR-035	Provide analytics information on how models are used.	Monitoring & Analytics Service	Tracks usage metrics.
FR-036	Provide metadata (usage count, success rate, creation time, training data size).	Model Registry, Monitoring & Analytics Service	Stores and reports metadata.

NFR-001–NFR-010	Performance, usability, maintainability, modifiability, flexibility.	Layered Architecture, Interfaces, Monitoring & Analytics, UI Layer	Overall structure and components chosen to satisfy quality attributes.
C-001–C-004	Constraints on data and optimization goals.	Churn Analysis Service, Recommendation Service, Optimization Service	Use historical attrition data, supervised learning, and dual optimization objectives.

6. Logical architectural components

6.1 User interface layer

Component Name: Model & Analytics UI **Primary Responsibility:** Provide interfaces for users to create/train models, load models, request monitoring information, and view reports. **Inputs:** User commands (model creation, loading, monitoring queries), filters, customer IDs. **Outputs:** Visual dashboards, forms, reports, status messages. **Public Interfaces:**

- **UI-CreateModel:** Start model creation workflow.
- **UI-LoadModel:** Upload/load model into system.
- **UI-MonitorModels:** View performance and optimization metrics.
- **UI-ViewReports:** Access model and churn reports. **Dependencies:** Model Management Service, Monitoring & Analytics Service, Model Reporting Module.

6.2 API gateway / client interaction layer

Component Name: Client API Gateway **Primary Responsibility:** Provide programming interfaces for client applications to request recommendations, churn predictions, and next best actions (FR-022–FR-027). **Inputs:** API requests (customer IDs, model IDs, action requests). **Outputs:** Ranked recommendations, churn scores, action sequences. **Public Interfaces:**

- **API-RecommendProducts(customerId)**
- **API-NextBestAction(customerId)**
- **API-ChurnPrediction(customerIds, modelId)** **Dependencies:** Recommendation Service, Action Recommendation Engine, Churn Prediction Service.

6.3 Data access and provisioning layer

Component Name: Data Access Layer **Primary Responsibility:** Abstract access to banking transaction datasets and other required data sources (dependencies in 2.2). **Inputs:** Queries from services (customer transactions, product usage, historical attrition). **Outputs:** Raw data sets, transaction streams. **Public Interfaces:**

- **DA-GetCustomerTransactions(customerId, timeRange)**
- **DA-GetHistoricalAttritionData()**

- **DA-GetProductUsage(customerId) Dependencies:** Banking transaction datasets (external).

Component Name: Data Provisioning Service **Primary Responsibility:** Provide daily batch data for training, scoring, and optimization (FR-003). **Inputs:** Requests for daily datasets, data from Data Access Layer. **Outputs:** Prepared daily datasets for models. **Public Interfaces:**

- **DP-GetDailyTrainingData(modelId)**
- **DP-GetDailyScoringData(modelId) Dependencies:** Data Access Layer, Data Preprocessing Service.

Component Name: Streaming Data Provisioning Service **Primary Responsibility:** Provide streaming data for training, scoring, and optimization (FR-004). **Inputs:** Continuous transaction events from Data Access Layer. **Outputs:** Streamed, preprocessed events to prediction/optimization services. **Public Interfaces:**

- **SD-SubscribeToCustomerEvents(modelId) Dependencies:** Data Access Layer, Data Preprocessing Service.

6.4 Data preprocessing layer

Component Name: Data Preprocessing Service **Primary Responsibility:** Perform normalization, outlier detection/extraction, missing data imputation, feature selection, and dimension reduction (FR-030–FR-031). **Inputs:** Raw datasets from Data Access Layer/Data Provisioning. **Outputs:** ML-ready datasets and feature sets. **Public Interfaces:**

- **DP-PreprocessDataset(datasetId)**
- **DP-ApplyFeatureSelection(datasetId) Dependencies:** Data Access Layer.

6.5 Customer profiling and tendency modeling

Component Name: Customer Profiling Service **Primary Responsibility:** Create and maintain customer profiles based on spending habits and product use behaviors (FR-001–FR-002). **Inputs:** Transaction and product usage data from Data Access Layer. **Outputs:** Customer profiles (segments, behavior descriptors). **Public Interfaces:**

- **CP-GenerateProfile(customerId)**
- **CP-GetProfile(customerId) Dependencies:** Data Access Layer, Data Preprocessing Service.

Component Name: Tendency Modeling Service **Primary Responsibility:** Provide customer tendency models including probability of purchasing/actively using specific banking products (FR-017). **Inputs:** Customer profiles, historical product usage. **Outputs:** Tendency scores per product. **Public Interfaces:**

- **TM-GetTendencyScores(customerId) Dependencies:** Customer Profiling Service, Model Management Service.

6.6 Model management and training

Component Name: Model Management Service **Primary Responsibility:** Manage ML models (creation, registration, lifecycle) (FR-007, FR-032, FR-019). **Inputs:** Model definitions, training configurations, user commands. **Outputs:** Registered models, status updates. **Public Interfaces:**

- **MM-CreateModel(config)**
- **MM-RegisterModel(modelArtifact)**
- **MM-UpdateModelStatus(modelId, status)** **Dependencies:** Model Registry, Training Service, Model Deployment Service.

Component Name: Model Registry **Primary Responsibility:** Store model metadata, status, and usage information (FR-015, FR-034–FR-036). **Inputs:** Model registration events, usage metrics. **Outputs:** Model lists, metadata. **Public Interfaces:**

- **MR-ListModels()**
- **MR-GetModelMetadata(modelId)** **Dependencies:** Monitoring & Analytics Service.

Component Name: Training Service **Primary Responsibility:** Train models, serialize/deserialize them (FR-011, FR-033, FR-018). **Inputs:** Preprocessed datasets, model configurations. **Outputs:** Trained model artifacts. **Public Interfaces:**

- **TR-TrainModel(modelId, datasetId)**
- **TR-SerializeModel(modelId)**
- **TR-DeserializeModel(modelArtifact)** **Dependencies:** Data Preprocessing Service, Model Registry.

Component Name: Model Deployment Service **Primary Responsibility:** Activate/deactivate models and implement them in production (FR-008, FR-033). **Inputs:** Model artifacts, activation commands. **Outputs:** Deployed models ready for prediction. **Public Interfaces:**

- **MD-ActivateModel(modelId)**
- **MD-DeactivateModel(modelId)** **Dependencies:** Model Registry, Prediction Services.

6.7 Prediction, churn analysis, and optimization

Component Name: Churn Analysis Service **Primary Responsibility:** Conduct churn analysis using historical data from customers who terminated their relationship (C-001, C-002, FR-026). **Inputs:** Historical attrition data, model outputs. **Outputs:** Analytical reports, churn factors. **Public Interfaces:**

- **CA-RunChurnAnalysis(modelId)** **Dependencies:** Data Access Layer, Churn Prediction Service.

Component Name: Churn Prediction Service **Primary Responsibility:** Provide churn prediction for given customer IDs and selected model (FR-027). **Inputs:** Customer IDs, selected model ID, current data. **Outputs:** Churn probabilities per customer. **Public Interfaces:**

- **CPR-PredictChurn(customerIds, modelId) Dependencies:** Model Registry, Data Access Layer, Data Preprocessing Service.

Component Name: Churn Ensemble Service **Primary Responsibility:** Utilize more than one model to conduct churn prediction (FR-028). **Inputs:** Multiple model IDs, customer data. **Outputs:** Combined churn predictions. **Public Interfaces:**

- **CE-PredictChurnEnsemble(customerIds, modelIds) Dependencies:** Churn Prediction Service, Model Registry.

Component Name: Recommendation Service **Primary Responsibility:** Provide ranked product/service recommendations for a given customer ID (FR-022–FR-023). **Inputs:** Customer profiles, tendency scores, optimization constraints. **Outputs:** Ranked recommendation lists. **Public Interfaces:**

- **RS-GetRecommendations(customerId) Dependencies:** Customer Profiling Service, Tendency Modeling Service, Optimization Service.

Component Name: Action Recommendation Engine **Primary Responsibility:** Determine best possible actions and sequenced actions to prevent attrition (FR-021, FR-024–FR-025). **Inputs:** Churn predictions, customer profiles, optimization results. **Outputs:** Next best action, ordered action sequences. **Public Interfaces:**

- **ARE-GetNextBestAction(customerId)**
- **ARE-GetActionSequence(customerId) Dependencies:** Churn Prediction Service, Optimization Service.

Component Name: Optimization Service **Primary Responsibility:** Provide optimization results for recommendations and actions, balancing company revenue expectations and customer satisfaction (FR-006, C-004). **Inputs:** Candidate recommendations/actions, business constraints. **Outputs:** Optimized recommendation/action sets. **Public Interfaces:**

- **OPT-OptimizeRecommendations(customerId)**
- **OPT-OptimizeActions(customerId) Dependencies:** Recommendation Service, Action Recommendation Engine, Monitoring & Analytics Service.

6.8 Monitoring, analytics, and reporting

Component Name: Monitoring & Analytics Service **Primary Responsibility:** Monitor model performance, optimization performance, and provide analytics on model usage (FR-010–FR-014, FR-035–FR-036, NFR-004–NFR-007). **Inputs:** Prediction outputs, actual results, optimization outcomes, usage events. **Outputs:** Metrics, dashboards, alerts, metadata. **Public Interfaces:**

- **MA-RecordPredictionOutcome(modelId, customerId, prediction, actual)**
- **MA-GetModelPerformance(modelId)**
- **MA-GetOptimizationPerformance() Dependencies:** Data Access Layer, Model Registry.

Component Name: Model Reporting Module **Primary Responsibility:** Provide reporting functions listing existing supervised learning models and their status (FR-029, FR-034). **Inputs:** Model metadata from Model Registry. **Outputs:** Reports on models in use. **Public Interfaces:**

- **MRP-GenerateModelReport()** **Dependencies:** Model Registry.

Component Name: Module Management Component **Primary Responsibility:** Manage newly added modules (FR-016). **Inputs:** Module registration requests. **Outputs:** Module registry entries, status. **Public Interfaces:**

- **MMC-RegisterModule(moduleDescriptor)** **Dependencies:** System configuration repository (logical).

7. Component interactions

7.1 Control flow

- **Model lifecycle:**
 1. User uses Model & Analytics UI to create a model (UI-CreateModel).
 2. UI calls Model Management Service (MM-CreateModel).
 3. Model Management Service requests training via Training Service (TR-TrainModel) using data from Data Provisioning and Data Preprocessing.
 4. Training Service serializes the model (TR-SerializeModel) and registers it in Model Registry (MR-ListModels/MR-GetModelMetadata).
 5. Model Deployment Service activates the model (MD-ActivateModel), making it available to Prediction and Recommendation services.
- **Prediction and churn analysis:**
 1. Client application calls Client API Gateway (API-ChurnPrediction).
 2. Gateway forwards request to Churn Prediction Service (CPR-PredictChurn).
 3. Churn Prediction Service fetches data via Data Access Layer and preprocesses it via Data Preprocessing Service.
 4. Predictions are generated using the selected model and recorded by Monitoring & Analytics Service (MA-RecordPredictionOutcome).
 5. Churn Analysis Service (CA-RunChurnAnalysis) may later use stored predictions and historical data for deeper analysis.
- **Recommendations and actions:**
 1. Client application calls API-RecommendProducts or API-NextBestAction.
 2. Gateway invokes Recommendation Service or Action Recommendation Engine.
 3. These components use Customer Profiling Service, Tendency Modeling Service, Churn Prediction Service, and Optimization Service to compute ranked recommendations and action sequences.
 4. Optimization Service ensures recommendations are optimized for revenue and satisfaction (C-004).
 5. Monitoring & Analytics Service records optimization performance for later reporting.

7.2 Data flow

- Transaction data flows from Banking transaction datasets through Data Access Layer to Data Provisioning/Streaming services, then to Data Preprocessing, and finally to Training/Prediction/Recommendation components.
- Model artifacts and metadata flow between Training Service, Model Deployment Service, Model Registry, and Monitoring & Analytics Service.
- Prediction outputs and actual results flow into Monitoring & Analytics Service for performance measurement and comparison (FR-013).

Communication between components is via logical interfaces (method calls or service calls), without assuming specific protocols.

8. Data management strategy

8.1 Data ownership

- **Customer transaction data:** Owned by external banking transaction datasets; accessed via Data Access Layer.
- **Customer profiles and tendency scores:** Owned by Customer Profiling Service and Tendency Modeling Service; stored in a logical profile store.
- **Model artifacts and metadata:** Owned by Model Registry; includes model status, usage count, success rate, creation time, training data size (FR-036).
- **Monitoring data (predictions vs actuals, optimization performance):** Owned by Monitoring & Analytics Service.

8.2 Persistence strategy

No specific database technology is assumed. Logical persistence responsibilities:

- Data Access Layer persists or retrieves transaction and attrition data from external banking datasets.
- Model Registry persists model metadata and artifacts.
- Monitoring & Analytics Service persists performance metrics and usage logs.
- Customer Profiling Service persists profiles and segments.

8.3 Data flow and storage responsibilities

- Preprocessed datasets are stored by Data Preprocessing Service for reuse in training and scoring.
- Historical attrition data is stored and accessed via Data Access Layer for churn analysis (C-001).
- Prediction results and actual outcomes are stored for comparison (FR-013) and performance measurement (FR-010, NFR-004).

9. External system interactions

Explicitly supported external systems per SRS:

- **Banking transaction datasets:** Source of customer transaction and product usage data (2.2 Dependencies).

- **Action services:** External services that execute recommended actions (e.g., retention campaigns), referenced as “Action services” in dependencies.
- **Client applications:** External applications consuming recommendations, churn predictions, and next best actions via programming interfaces (FR-022).

No other external systems (authentication, payments, notifications) are introduced.

10. Architectural quality considerations

10.1 Performance

- Separation of training and prediction services ensures online prediction is not blocked by heavy training workloads (NFR-001).
- Data Provisioning and Streaming services provide appropriate data flows for timely scoring and optimization.

10.2 Usability

- Model & Analytics UI provides dedicated interfaces for model creation, loading, and monitoring (NFR-002–NFR-003).
- Clear APIs via Client API Gateway simplify integration for client applications (FR-022).

10.3 Maintainability

- Layered architecture and modular components (Data Preprocessing, Model Management, Monitoring) support easy maintenance and evolution (NFR-004–NFR-007).
- Monitoring & Analytics Service centralizes performance measurement and comparison of predictions vs actuals.

10.4 Modifiability and flexibility

- Interfaces between layers allow underlying technologies (libraries, languages) to change without affecting higher-level logic (NFR-008–NFR-010).
- Model Registry and Module Management Component support adding new models and modules without structural changes.

11. Architectural decisions and trade-offs

Decision	Rationale	Advantages	Potential Disadvantages
Use layered, modular architecture	Aligns with maintainability, flexibility, and clear separation of concerns.	Easier evolution, clear responsibilities, supports NFR-004–NFR-010.	May require more coordination between layers; potential overhead.

Separate training and prediction services	NFR-005 requires same system for both but performance demands separation.	Online prediction remains responsive; training can be scheduled.	More complex deployment and monitoring of multiple services.
Introduce dedicated Monitoring & Analytics Service	NFR-004–NFR-007 emphasize performance measurement and comparison.	Centralized metrics, easier reporting and analysis.	Additional component to maintain; requires consistent data collection.
Use Model Registry as central metadata store	FR-015, FR-034–FR-036 require model lists and metadata.	Single source of truth for models and their status.	Registry becomes critical; must be highly reliable.
Provide Client API Gateway	FR-022 requires programming interface for client applications.	Simplifies external integration, enforces consistent APIs.	Gateway may become a bottleneck if not scaled appropriately.

12. Architectural risks and limitations

- **Data quality risk:** Churn prediction and recommendations depend heavily on the quality and completeness of transaction data; poor data may reduce effectiveness.
- **Model management complexity:** Managing many models (including ensembles) with metadata and performance tracking can become complex and may require careful governance.
- **Performance under high load:** Streaming data and real-time predictions may challenge performance if not adequately scaled, though scaling strategies are outside SRS scope.
- **Ambiguity in “Action services”:** The SRS mentions action services but does not specify their capabilities; integration assumptions may need refinement.

Areas requiring future refinement:

- Detailed data schemas for transaction and profile data.
- Specific optimisation criteria and constraints beyond “revenue expectations and customer satisfaction” (C-004).
- Detailed user roles and permissions for UI and APIs.

13. Mermaid component diagram

